

Supporting Information for

Dissection of 3D chromosome organization in *Streptomyces coelicolor* A3(2) leads to biosynthetic gene cluster overexpression

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Materials and Methods

Bacterial strains and growth conditions

S. coelicolor M145 strain (specimen voucher: ATCC BAA-471) and mutant strains were grown on solid soy flour-mannitol (SFM) medium (20 g/L mannitol, 33 g/L soya flour, 20 g/L agar) at 28°C for sporulation. Then the spores stocked in 20% glycerol were inoculated to 2×TY medium (16 g/L tryptone, 10 g/L yeast extract, 5 g/L NaCl) and germinating in rotary incubator with 30°C and 220 revolutions per minute (rpm) for 8 h. The total cells were collected and inoculated to liquid R5- medium containing stainless steel coil and grow about 14 h to M phase ($OD_{450} = 0.5 \sim 0.7$) or 22 h to L phase ($OD_{450} = 1.5 \sim 2.0$) in rotary incubator with 30°C and 220 rpm.

Composition of the liquid R5- medium is as following: 25 mM TES (pH 7.2), 103 g/L sucrose, 10 g/L glucose, 5 g/L yeast extract, 10.12 g/L $MgCl_2 \cdot 6H_2O$, 0.25 g/L K_2SO_4 , 0.1 g/L casamino acids, 0.08 mg/L $ZnCl_2$, 0.4 mg/L $FeCl_3 \cdot 6H_2O$, 0.02 mg/L $CuCl_2 \cdot 2H_2O$, 0.02 mg/L $MnCl_2 \cdot 4H_2O$, 0.02 mg/L $Na_2B_4O_7 \cdot 10H_2O$ and 0.02 mg/L $(NH_4)_6Mo_7O_{24} \cdot 4H_2O$ and adjusted pH to 6.8 with 5 M NaOH. The total cells of different samples and growth phase were harvested and proceeded to further investigation. *E. coli* DH10B was used as the cloning host and *E. coli* ET12567/pUZ8002 was used for intergeneric conjugation. All *E. coli* strains were cultured in 2×TY medium at 37°C.

The construction of plasmid and *Streptomyces* mutant

S. coelicolor strains, *E. coli* strains, plasmids and designed sgRNA have been summarized in Tables S2–5. DNA manipulations were performed by standard protocols (1).

For plasmids construction, different sgRNA targeted to each HCR-MN and HCR-LN windows were one-step clone to *NheI* and *XbaI* digested *Streptomyces* code-optimized *spCas9* born plasmid pWHU2653 with promoter *kasO***p* and B1006 terminator by Gibson assembly. Then a series of left and right homologous arms together with the *gus(a)* reporter cassette or the RK-682 recombinant BGC cassette were introduced to the *HindIII* site of these sgRNA born plasmids by Gibson assembly to achieve plasmids pGAM1 ~ 10, pGAL1 ~ 10, pRKM1 ~ 10 and pRKL1 ~ 10, respectively. The *gus(a)* reporter cassette or RK-682 recombinant BGC cassette was integrated into *PvuI* and *XbaI* linearized pSET152 to produce pGAattB or pRKattB, respectively.

For *S. coelicolor* mutants construction, each plasmid with pWHU2653 backbone was transformed from *E. coli* ET12567/pUZ8002 to *S. coelicolor* by intergeneric conjugation with standard protocols (2). The apramycin-resistant exconjugants were patched on SFM plate with 25 µg/mL apramycin and grown at 28°C for 2 or 3 days to confirm their antibiotic resistance and then transferred to a SFM plate without antibiotics using plate streaking. The single clones were then patched to SFM plates with 25 µg/mL apramycin and SFM plates without antibiotic, respectively. To screen for the recombinant mutant, the candidate mutants with apramycin-sensitive phenotype were further verified by PCR and sequencing. Two plasmids with pSET152 backbone were transferred to *S. coelicolor* by intergeneric conjugation and the apramycin-resistant exconjugants were also verified by PCR and sequencing.

Measurement of glucuronidase activity

For the glucuronidase activity test, *S. coelicolor* wild-type and mutant strains were cultured in liquid R5- medium as previous description. After reaching the specific growth phase, the whole cells were harvested by centrifugation and resuspended with 20 mL 40% glycerol. 15 mL of the cell suspension was separated and centrifugated, then washed twice with distilled water. This part was then evaporated to dryness under reduced pressure and then measure the dry weight. The rest 5 mL cell suspension was also centrifugated, the supernatant discarded and then resuspended with 10 mL GUS buffer (50 mM phosphate buffer, pH 7.0; 5 mM DTT; 0.1% Triton X-100). The cell suspension was then lysed by sonication. 200 μ L cell lysate with six repeats was transferred to 96-well plates and then supplemented with 5 μ L 100 mM *p*-nitrophenyl- β -D-glucuronide. Incubating at 37°C, the optical density at 415 nm was measured per 5 min for a total of 40 minutes. Glucuronidase activity (U/g) in our case was calculated as: ΔOD_{415} (0 min to 40 min)/dry weight (g).

Isolation, detection, and quantification of RK-682

For RK-682 production and analysis, *S. coelicolor* wild-type and mutant strains were cultured in liquid R5- medium as the previous description. After reaching the specific growth phase, the whole cells were harvested by centrifugation and then washed twice with distilled water. These cells were evaporated to dryness under reduced pressure and then measure the dry weight. These dry cells were extracted with 40 mL ethyl acetate and the extracts were then evaporated to dryness and redissolved in 1 mL methanol for further analysis. Analysis of RK-682 was performed with a Phenomenex C18 column (5 μ m, 4.6 $\times$ 250 mm) with elution at a flow rate of 1 mL/min over a 22 min gradient as follows: time T = 0, 85% B; T = 14 min, 95% B; T = 18 min, isocratic gradient with 95% B; T = 19 min, 85% B; T = 22 min, isocratic gradient with 85% B (phase A, 20 mM ammonium acetate; phase B, methanol). LC-ESI-HRMS analysis was carried out on an LTQ XL Orbitrap (Thermo Fisher Scientific) coupled with an Accela photodiode array (PDA) detector, Accela PDA autosampler, and Accela pump, using electrospray ionization in the negative-ion mode. The selective ion monitoring was carried out on $[M+H]^-$ (*m/z*, 367.2479) ions of RK-682 and the area of extracted peaks was counted. Relative RK-682 production (U/mg) in our case was calculated as: peaks-area/dry weight (mg).

Statistical information

The Microsoft Excel software package and Graphpad Prism software were used for statistical calculations and data visualization. All data presented are means $\pm$ standard errors calculated from three independent experiments.

RNA extraction

Total RNA was extracted from the tissue using TRIzol[®] Reagent according to the manufacturer's instructions (Invitrogen) and genomic DNA was removed using DNase I (TaKara). Then RNA quality was determined by 2100 Bioanalyser (Agilent) and quantified using the ND-2000 (NanoDrop Technologies). Only high-quality RNA sample ($OD_{260}/OD_{280} = 1.8\sim 2.0$, $OD_{260}/OD_{230} \geq 2.0$, RIN ≥ 6.5 , 28S : 18S ≥ 1.0 , concentration ≥ 100 ng/ μ L, mass ≥ 2 μ g) was used to construct sequencing library.

Library construction and sequencing

RNA-seq transcriptome library was prepared following TruSeq™ RNA sample preparation Kit from Illumina (San Diego) using 2 µg of total RNA. Shortly, ribosomal RNA (rRNA) depletion instead of poly(A) purification is performed by Ribo-Zero Magnetic Kit (Epicenter) and then all mRNAs were broken into short (200 nt) fragments by adding fragmentation buffer firstly. Secondly double-stranded cDNA was synthesized using a SuperScript double-stranded cDNA synthesis Kit (Invitrogen) with random hexamer primers (Illumina). When the second strand cDNA was synthesized, dUTP was incorporated in place of dTTP. Then the synthesized cDNA was subjected to end-repair, phosphorylation, and 'A' base addition according to Illumina's library construction protocol. The second strand cDNA with dUTP was recognized and degraded by the UNG enzyme. Libraries were size selected for cDNA target fragments of 200 bp on 2% Low Range Ultra Agarose followed by PCR amplified using Phusion DNA polymerase (NEB) for 15 PCR cycles. After quantified by TBS380, the paired-end RNA-seq sequencing library was sequenced with the Illumina HiSeq × TEN (2 × 150 bp read length). The processing of original images to sequences, base-calling, and quality value calculations were performed using the Illumina GA Pipeline (version 1.6), in which 150 bp paired-end reads were obtained. A Perl program was written to select clean reads by removing low-quality sequences, reads with more than 5% of N bases (unknown bases) and reads containing adaptor sequences.

RNA-seq analysis

htqc (3) (version 0.90.8) was used to filter the sequencing data to make clean data. Then the cleaned reads were mapped onto the *S. coelicolor* genome (NC_003888.3) using SOAP (4) (2.2.1). The significant differentially expressed genes (DEGs) were identified by edgeR (5) with definition of log2 fold change (log2FC) >1 or < -1 and false discovery rate (FDR) <0.05.

Hi-C library construction

Hi-C libraries were constructed according to previous studies (6). Briefly, bacterial cells were cross-linked with 3% formaldehyde for 1 hour at 4°C and quenched with 375 mM final concentration glycine for 15 min. The cross-linked cells were incubated with lysozyme at 37°C for 10 min to lyse the cells. Endogenous nucleases were inactivated with 0.5% SDS, then chromatin DNAs were digested by 50 U Sau3AI (NEB) and marked with biotin-14-dCTP (Invitrogen) and then ligated by 50 U T4 DNA ligase (NEB). After reversing cross-links, the ligated DNA was extracted using the QIAamp DNA Mini Kit (Qiagen) according to the manufacturers' instructions. Purified DNA was sheared to 300 ~ 500 bp fragments, then blunt-end repaired, A-tailed and adaptor-added before purification through biotin-streptavidin-mediated pull-down and PCR amplification. Finally, the Hi-C libraries were quantified and sequenced on the Illumina Nova-seq platform (San Diego).

Construction of contact map

After quality filtering using htqc (3) (version 0.90.8), the clean Hi-C data from two biological replicates for M phase and L phase, was iteratively mapped to the *S. coelicolor* genome (NC_003888.3) using the hiclib (7) software package (version 1f8815d0cc9e). Dangling ends and other unusable data were filtered, the valid pairs were used to analyze the correlation efficiency of the two biological replicates for each sample using QuASAR-Rep analysis (8) (3DChromatin_ReplicateQC v 0.0.1), then we pooled the data from the two replicates together

for further analysis. The valid pairs after pooling were binned into 40 kb, 5 kb and 2 kb nonoverlapping genomic intervals to generate contact maps. The contact maps were normalized using an iterative normalization method to eliminate systematic biases.

Compartment A/B analysis

The CscoreTool (v1.1) (9) software was applied to identify A and B compartments on 40 kb resolution. Genomic bins with a positive or negative c-score signa were divided into the A or B compartment. The active “A” compartments are active gene-enriched regions, whereas the inactive “B” compartments are inactive gene-enriched regions.

Chromosomal interaction domains (CIDs) analysis

CIDs are contiguous regions that display high levels of self-association and that are separated from adjacent regions by distinct boundaries. The locations of CIDs were determined at 5 kb resolution using Tung B. K. Le’s methods (10). For each bin, interactions between that bin and bins upstream or downstream within 200 kb were extracted to form two vectors. *T*-values of paired test of Log2 values of the two vectors were calculated to form a directional index (DI). A CID boundary is determined as the DI value upstream a bin is below -1.81 and downstream above 1.81 (*T*-value at which *P*-value = 0.05, degree of freedom = 10).

Calculation of significant interactions

The contacts between 5 kb bins of interactions of each sample were transferred to Ay’s Fit-Hi-C software (v2.0.0) (11) (with parameter settings-L 5000 - p 2) to calculate the corresponding cumulative probability *P*-value and *Q*-value. After calculation, the interactions in which both the *P*-value and *Q*-value were less than 0.01 and the contact count > 2 were identified as significant interactions.

Calculation of the FIRE value

The FIRE value was calculated referring to Schmitt A. D.’s methods (12). First, the genome effective fragment length and GC represent content and mappability (F_GC_MAP) were computed using the F_GC_MAP.file.sh script from the HiCnv software (GitHub - ay-lab/HiCnv: HiCnv is used to call copy number variations and breakpoints from Hi-C data). Second, the local interactions (contact sum of distance range of 15 ~ 200 kb for each bin from 5 kb raw matrix) were counted. Third, the HiCNormCis algorithm is the same as HiCNorm (13). They are both Poisson-regression-based normalization but HiCNorm is work on 2D data while HiCNormCis takes on 1D data. We modified the HiCNorm software code to take HiCNormCis calculation using the local interactions and F_GC_MAP of each bin. The code file was appended as [SI dataset 5](#).

Comparison of FIRE-like signals with RNA-seq

FIRE-like signals were calculated using Schmitt A. D.’s methods (12). RNA-seq coverage signals and FIRE-like signals were z-transformed and Gaussian Smoothed using Virginia S’ methods (2) to measure the correlation between transcription and local contact.

Construction of the chromosomal 3D structure

Pastis (v0.1) (14) was used to construct the chromosomal 3D structures using a 5 kb contact map with a multidimensional scaling (MDS) model.

Data availability

The genome sequence of *S. coelicolor* A3(2) was obtained from NCBI (Reference Sequence ID: NC_003888.3). The RNA-seq and Hi-C seq data generated during this study have been deposited in the NCBI BioProject under the accession code: PRJNA891940.

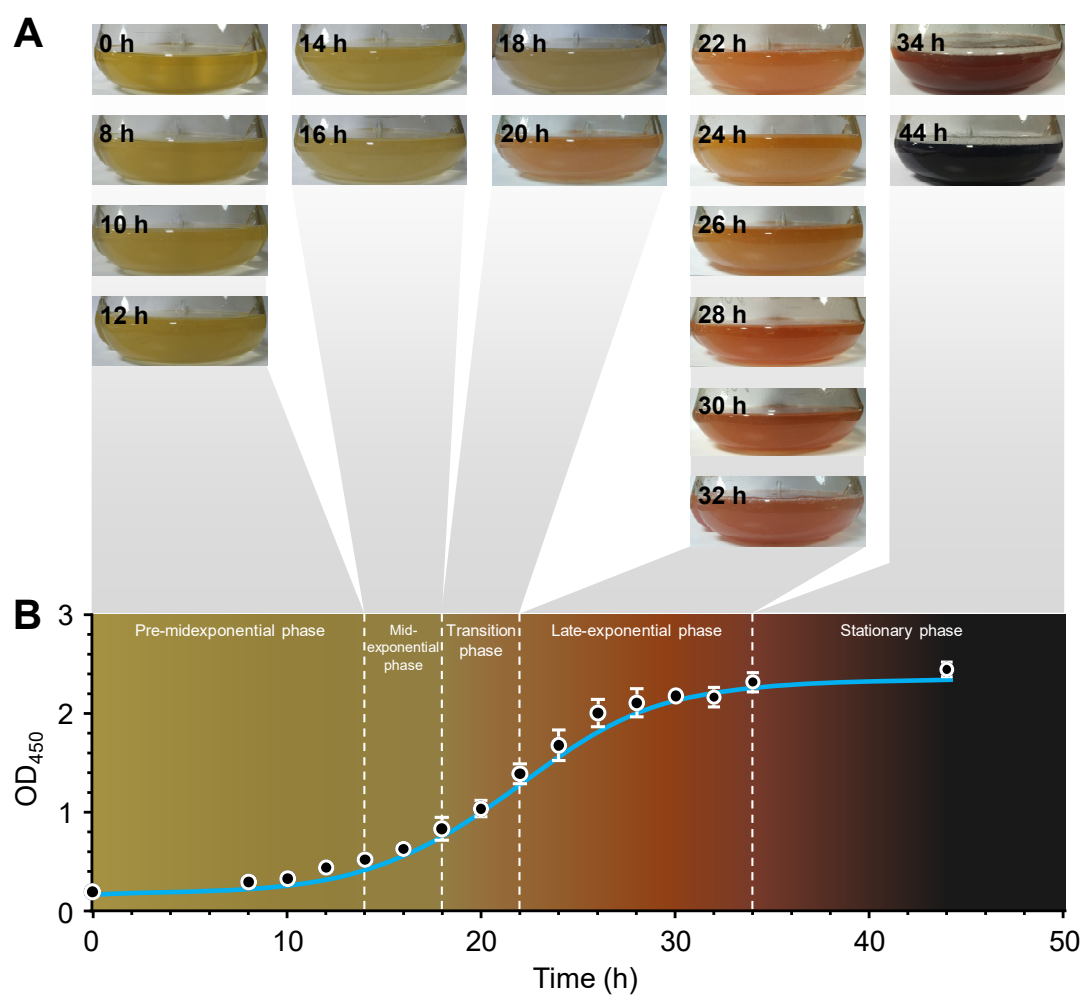


Fig. S1. Fermentation of *S. coelicolor* grown in R5- medium. (A) Fermentation broth in different phases. (B) Growth curve. The color of the background corresponds to the authentic broth.

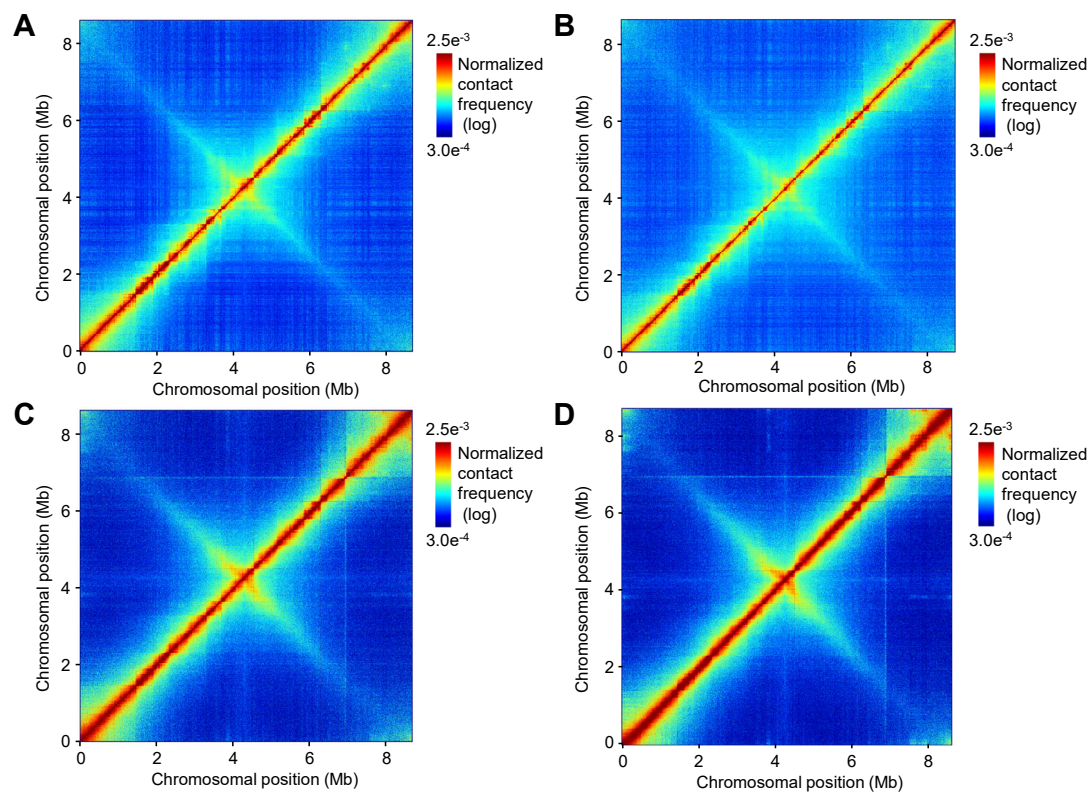


Fig. S2. The genomic contact map of two independent biological repeats in M phase (A and B, GenomeDISCO = 0.941) and L phase (C and D, GenomeDISCO = 0.981).

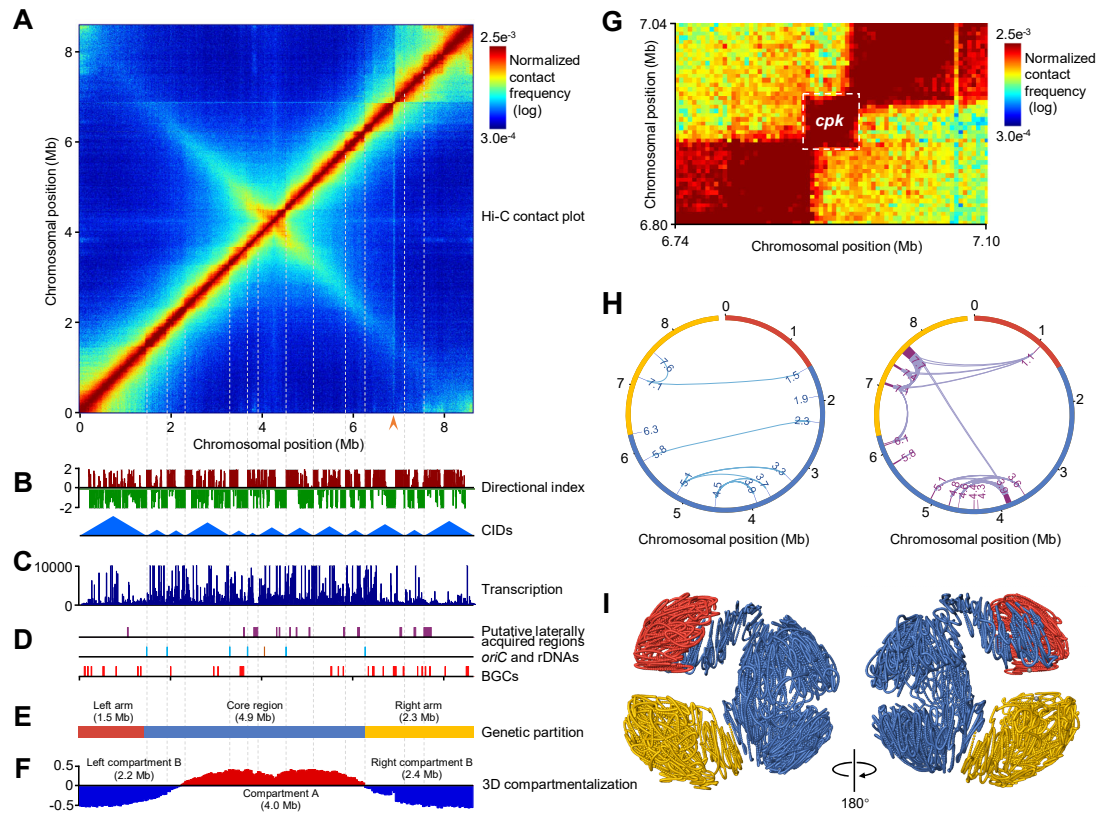


Fig. S3. Chromosome organization of *S. coelicolor* in L phase. (A) Normalized genomic contact map obtained from asynchronous population (22 h for L phase, R5- medium, 5 kb resolution). The orange arrow indicates the region interacted with the whole chromosome (B) Directional index analysis and identified CIDs (blue triangles). Dotted grey vertical lines crossing A to F highlight the barriers between CIDs. (C) RNA-seq profile mapped onto *S. coelicolor* genome in 1 kb resolution. (D) Genomic annotations as Fig. 1D. (E) Genetic partition in *S. coelicolor* as Fig. 1E. (F) 3D compartmentalization in *S. coelicolor*. The red and blue biases represent the A and B compartments in L phase, respectively. (G) The local chromosomal domain in *cpk* BGC loci. The independent interactive domain is highlighted by a dotted white frame. (H) Circos plots show the significant interaction among 12 CID boundaries (blue rectangle, left panel) or 14 putative laterally acquired regions (purple rectangle, right panel). Each connection/arc represents a significantly enriched interaction. (I) 3D representation of the contact map from A.

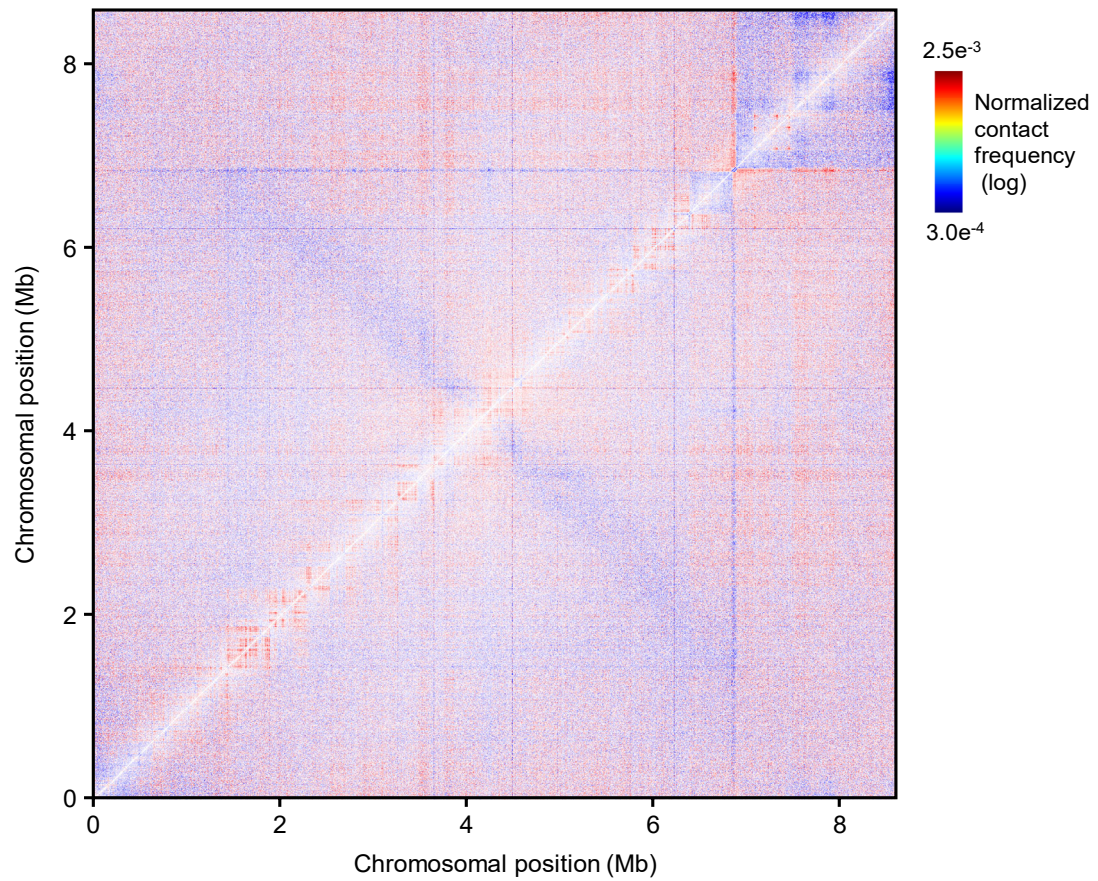


Fig. S4. The different Hi-C map in the 2 kb resolution comparing the contact enrichment in M phase versus L phase. the color scale reflects the different contact frequencies in two growth phases, from dark blue (frequent contacts in L phase) to dark red (frequent contacts in M phase).

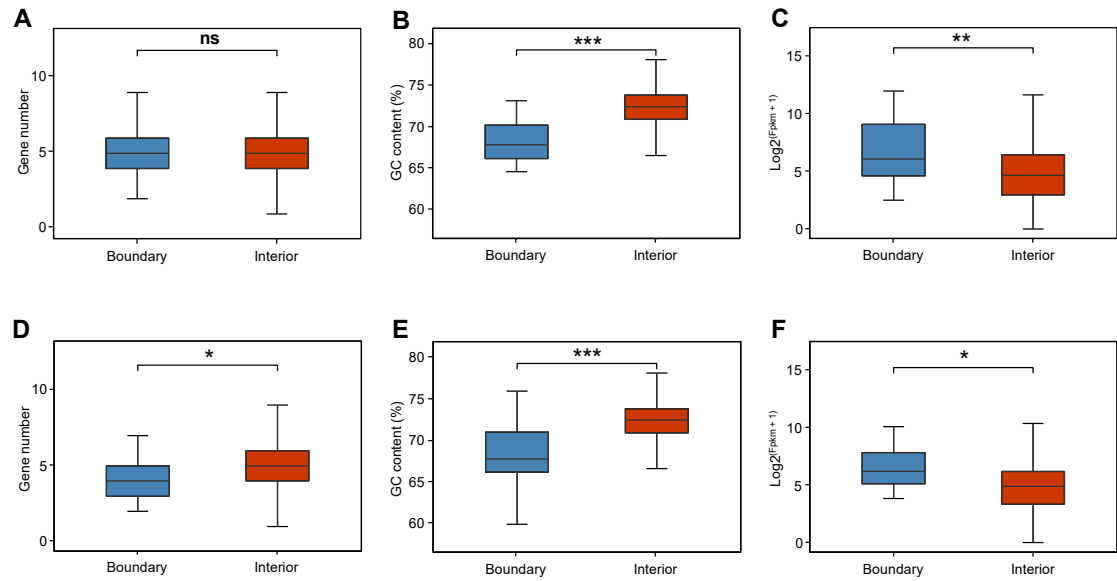


Fig. S5. The boxplots represent the comparison of gene density, GC content and expression between genes located in CID boundary or CID interior. (A) The comparison of gene density in M phase. (B) The comparison of GC content in M phase. (C) The comparison of gene expression in M phase. (D) The comparison of gene density in L phase. (E) The comparison of GC content in L phase. (F) The comparison of gene expression in L phase. Boxplots show median with first and third quartile while the lower and upper 'whiskers' extend to the value no further than $1.5 \times \text{IQR}$ from the 'hinge'. The statistical significance between boundary and interior determined by two-sided Wilcoxon rank sum test is marked with asterisks: P -value ≤ 0.05 (*), ≤ 0.01 (**), ≤ 0.001 (***) and no significance (ns).

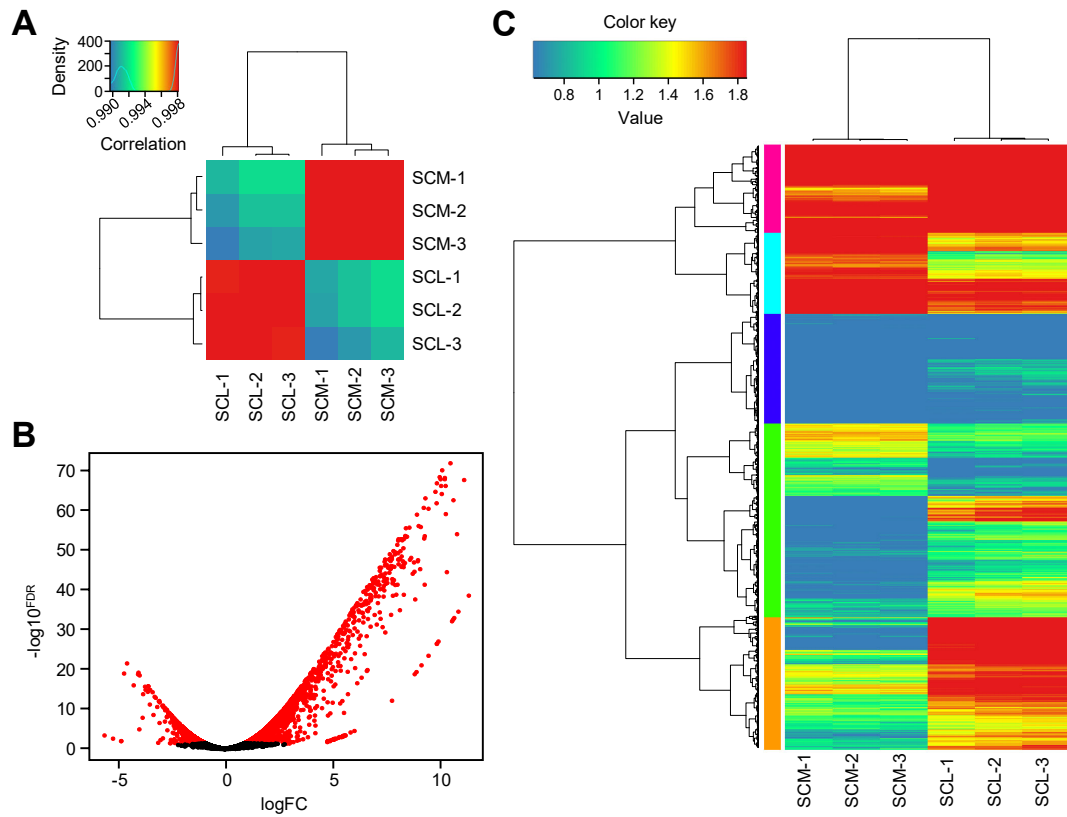


Fig. S6. Gene expression difference between M phase and L phase. (A) The Correlations between each three samples in two growth phases. The RNA-seq results are highly reproducible. (B) The volcano plot for differentially expressed genes (DEGs) in two growth phases. The horizontal axis represents the logFC (Fold Change), and the vertical axis represents the adjusted $-\log_{10}^{FDR}$ (False Discovery Rate). The red dots indicate DEGs ($\log_{FC} > 1$ or $\log_{FC} < -1$ and adjusted $-\log_{10}^{FDR} < 0.05$). (C) Heat map represent different expressed genes between the two growth phases.

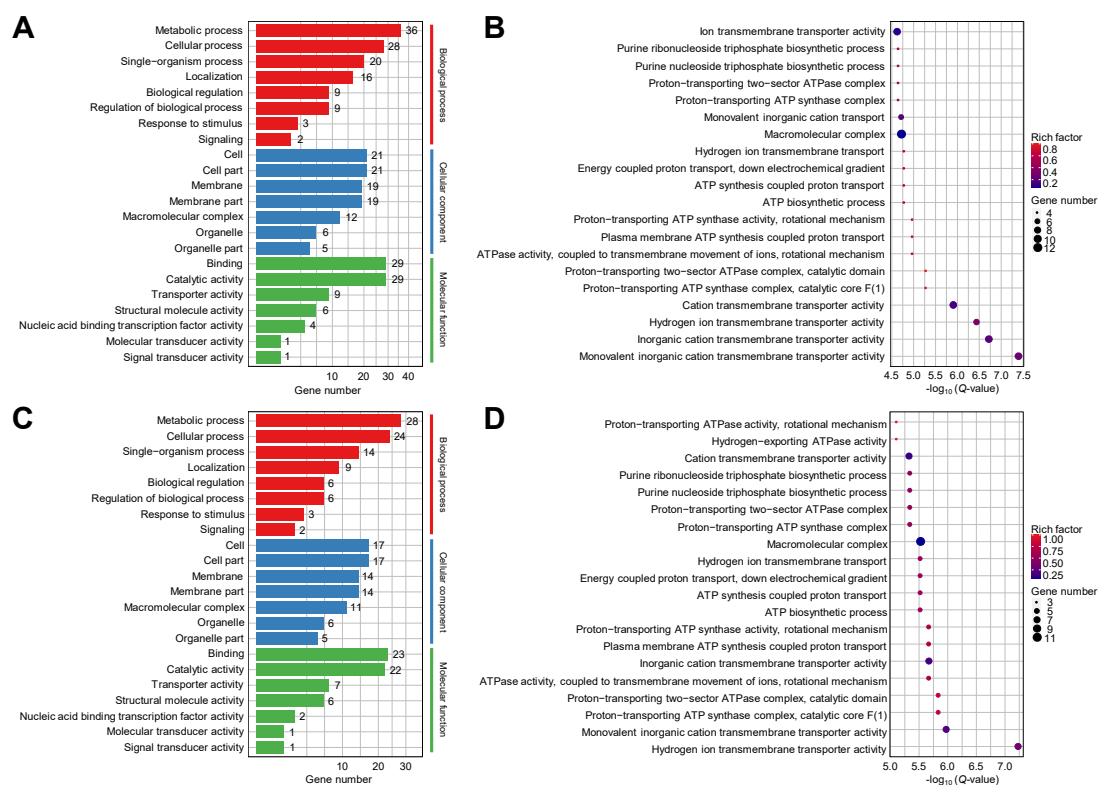


Fig. S7. Comparison of classification and enrichment gene function of CID boundaries genes in M phase and L phase (Gene Ontology, see method). (A) Classification of CID boundaries genes in M phase. (B) Enrichment of CID boundaries genes in M phase. (C) Classification of CID boundaries genes in L phase. (D) Enrichment of CID boundaries genes in L phase.

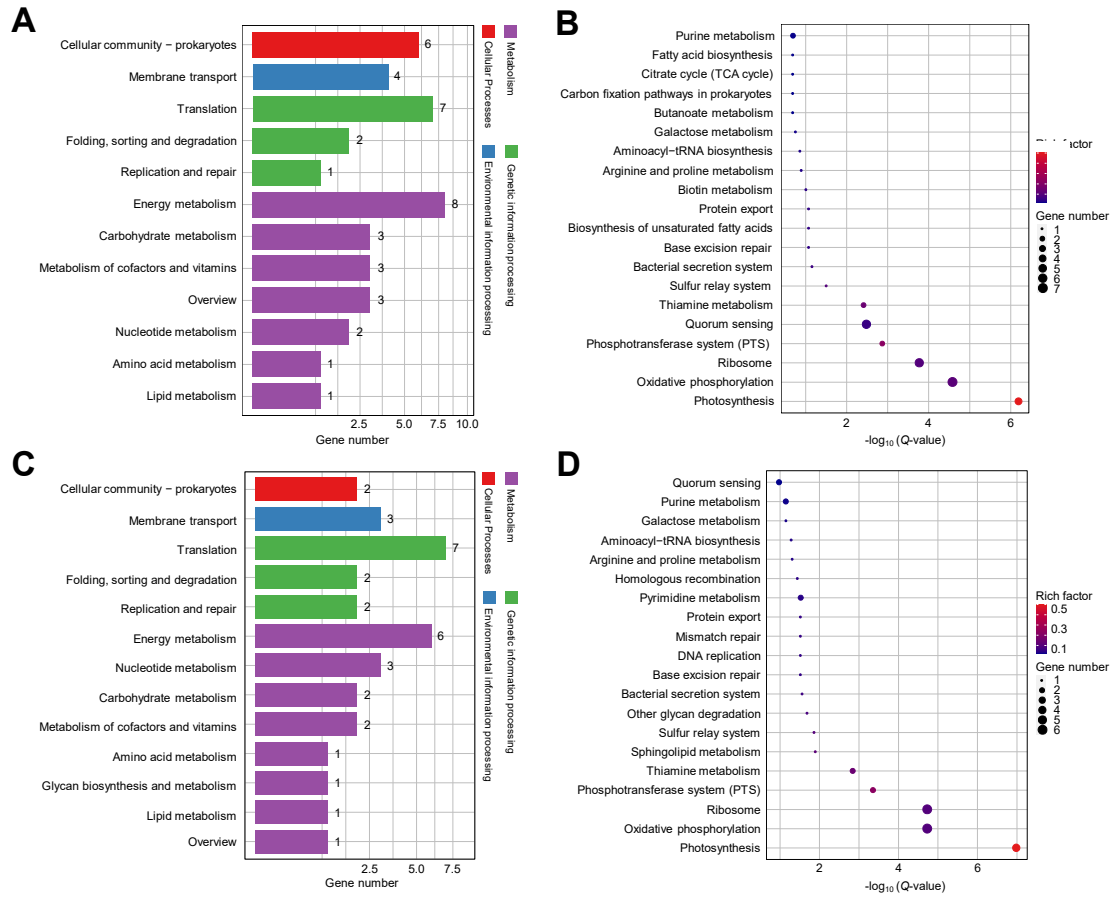


Fig. S8. Comparison of classification and enrichment pathways of CID boundaries genes in M phase and L phase (KEGG, see Materials and Method). (A) Classification of CID boundaries genes in M phase. (B) Enrichment of CID boundaries genes in M phase. (C) Classification of CID boundaries genes in L phase. (D) Enrichment of CID boundaries genes in L phase.

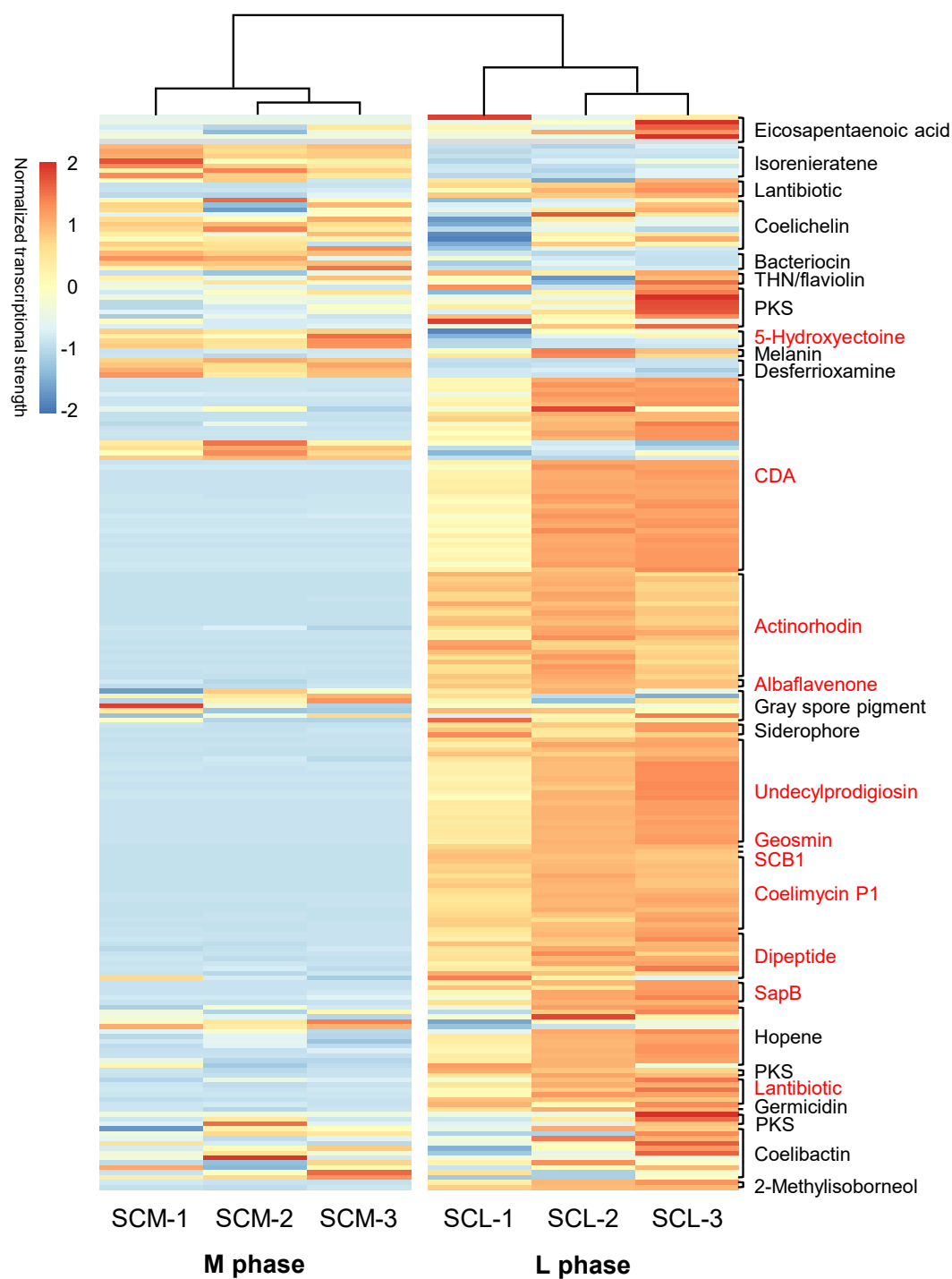


Fig. S9. Comparison of transcription of 28 BGCs in M phase and L phase. The color scale reflects the normalized transcriptional strength, from dark blue (rare expression) to dark red (high expression). The 11 BGCs with significant different expression in the two growth phases are highlighted in red font.

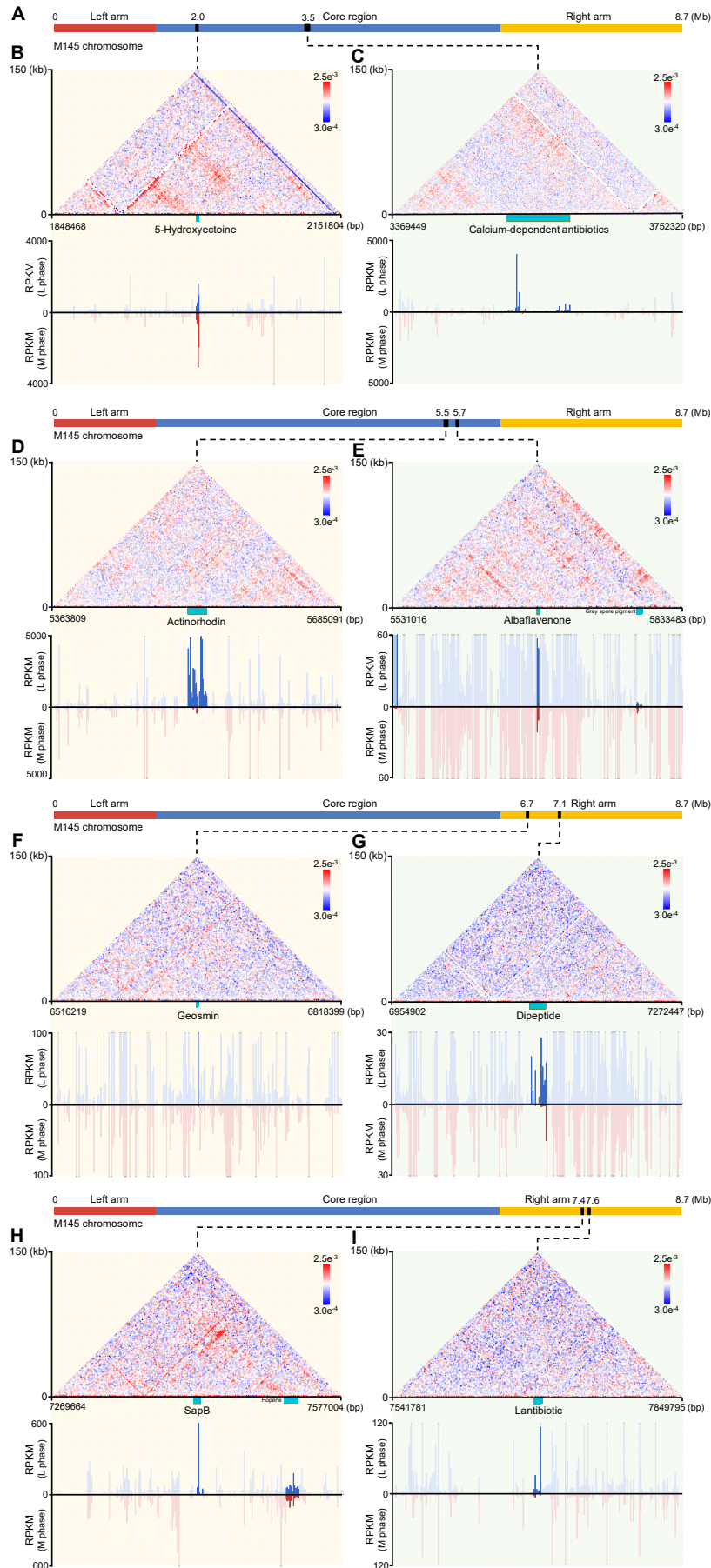


Fig. S10. The coordination between local chromosomal organization and transcription of BGCs. (A) The map of *S. coelicolor* chromosome with the same genetic partition as Fig. 1A. The relative location of dissected BGCs is represented by black rectangle. (B - I), Top panel: The different local Hi-C maps in the 2 kb resolution comparing the contact enrichment in M phase versus L phase, the exhibited range comprising the 5-hydroxyectoine (B), calcium-dependent antibiotics (C), actinorhodin (D), albaflavonone (E), geosmin (F), dipeptide (G), sapB (H), lantibiotic (I) BGC and each 150 kb flanking, respectively. the color scale reflects the different contact frequency in two growth phases, from dark blue (frequent contacts in L phase) to dark red (frequent contacts in M phase). Down panel: The normalized RNA-seq profile of single gene mapped onto the same *S. coelicolor* genome scope as each top panel. The RNA-seq profiles (reads per kilobase per million mapped reads, RPKM) in two growth phases are exhibited in mirror symmetry.

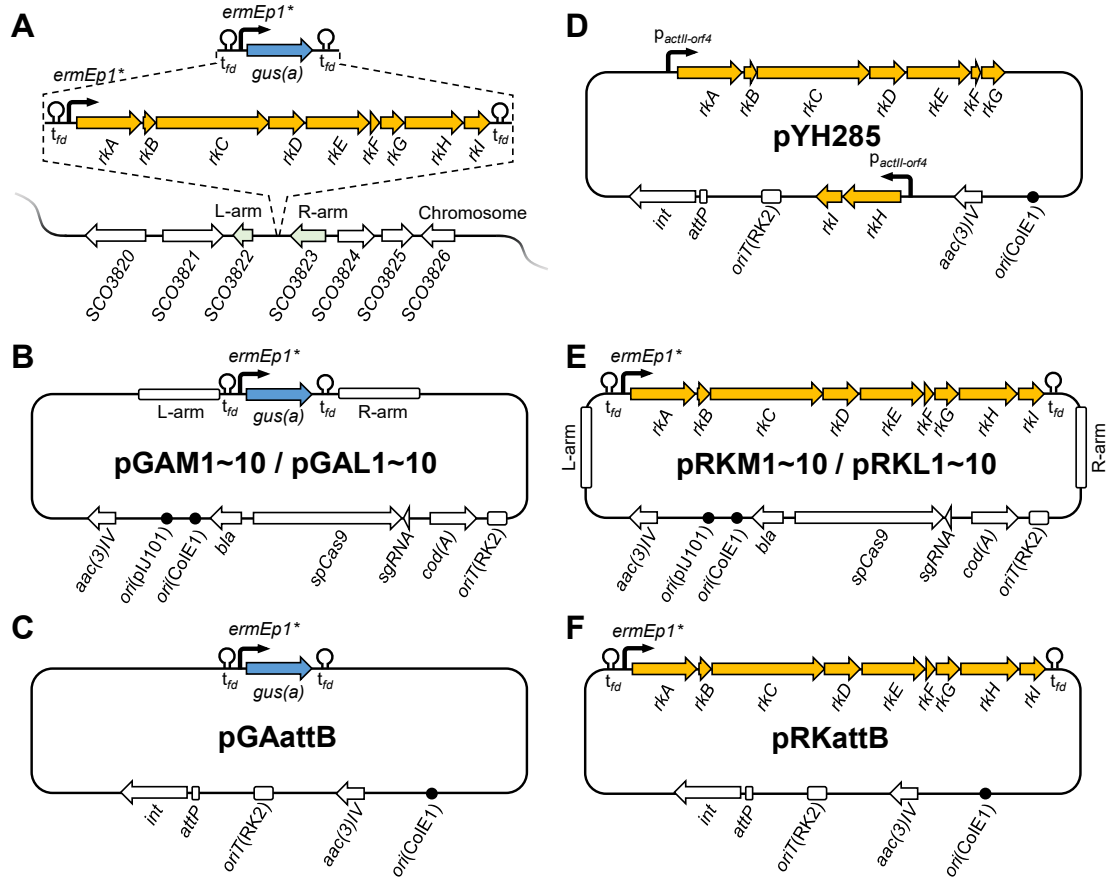


Fig. S11. Reporter gene and RK-682 BGC cassette used for integration in the chromosome. (A) Schematic representation of the target integration strategy (HCR-L1 site as the example). To minimize the potential interference between insert cassette and endogenous genes, the two kinds of cassette were preferred to integrate between two operons with the same transcriptional direction (arrowheads in light green) in the opposite orientation. (B) CRISPR/Cas9 based plasmids comprising *gus(a)* gene under the control of a constitutive promoter *ermEp1**, sandwiched by two bidirectional *fd*-phage terminators (*t_{fd}*). A series of homologous arms and sgRNAs were designed for different target integration of *gus(a)* report cassette (Table S4). (C) Integrative plasmid target to *attB* site with the same *gus(a)* report cassette as B. (D) Heterologous expression plasmid with entire RK-682 BGC under the control of two individual endogenous promoters (*P_{actII-orf4}*). (E) CRISPR/Cas9 based plasmids comprising fused entire RK-682 BGC under the control of one constitutive promoter *ermEp1**, sandwiched by two bidirectional *fd*-phage terminators. A series of homologous arms and sgRNAs were designed for different target integration of RK-682 BGC cassette (Table S4). (F) Integrative plasmid target to *attB* site with the same RK-682 BGC cassette as E.

Table S1. HCRs information in the study

Name	Start (nt)	End (nt)	PC	Fire value
HCR-M1	4205000	4260000	0.803	2.525
HCR-M2	4360000	4415000	0.829	2.283
HCR-M3	5055000	5110000	0.854	1.716
HCR-M4	3450000	3505000	0.938	1.566
HCR-M5	2790000	2845000	0.867	1.321
HCR-M6	6075000	6130000	0.823	1.097
HCR-M7	1995000	2050000	0.836	0.839
HCR-M8	860000	915000	0.827	0.605
HCR-M9	8120000	8175000	0.805	0.442
HCR-M10	8615000	8667507	0.963	0.119
HCR-L1	4190000	4245000	0.844	1.769
HCR-L2	4395000	4450000	0.836	1.573
HCR-L3	5030000	5085000	0.878	1.453
HCR-L4	4870000	4925000	0.816	1.345
HCR-L5	6055000	6110000	0.805	1.131
HCR-L6	1985000	2040000	0.802	0.966
HCR-L7	6935000	6990000	0.805	0.816
HCR-L8	8395000	8450000	0.807	0.623
HCR-L9	15000	70000	0.843	0.326
HCR-L10	8615000	8667507	0.968	0.175

Table S2. *S. coelicolor* strains used in the study

Name	Relevant genotype and characteristics	Source
M145 (wild-type)	<i>S. coelicolor</i> A3(2) SCP1-, SCP2-	(2)
gaM1	M145:: <i>gus(a)</i> in HCR-M1	This study
gaM2	M145:: <i>gus(a)</i> in HCR-M2	This study
gaM3	M145:: <i>gus(a)</i> in HCR-M3	This study
gaM4	M145:: <i>gus(a)</i> in HCR-M4	This study
gaM5	M145:: <i>gus(a)</i> in HCR-M5	This study
gaM6	M145:: <i>gus(a)</i> in HCR-M6	This study
gaM7	M145:: <i>gus(a)</i> in HCR-M7	This study
gaM8	M145:: <i>gus(a)</i> in HCR-M8	This study
gaM9	M145:: <i>gus(a)</i> in HCR-M9	This study
gaM10	M145:: <i>gus(a)</i> in HCR-M10	This study
gaL1	M145:: <i>gus(a)</i> in HCR-L1	This study
gaL2	M145:: <i>gus(a)</i> in HCR-L2	This study
gaL3	M145:: <i>gus(a)</i> in HCR-L3	This study
gaL4	M145:: <i>gus(a)</i> in HCR-L4	This study

gaL5	M145:: <i>gus(a)</i> in HCR-L5	This study
gaL6	M145:: <i>gus(a)</i> in HCR-L6	This study
gaL7	M145:: <i>gus(a)</i> in HCR-L7	This study
gaL8	M145:: <i>gus(a)</i> in HCR-L8	This study
gaL9	M145:: <i>gus(a)</i> in HCR-L9	This study
gaL10	M145:: <i>gus(a)</i> in HCR-L10	This study
gaAttB	M145:: <i>gus(a)</i>	This study
rkM1	M145:: <i>rk-682</i> in HCR-M1	This study
rkM2	M145:: <i>rk-682</i> in HCR-M2	This study
rkM3	M145:: <i>rk-682</i> in HCR-M3	This study
rkM4	M145:: <i>rk-682</i> in HCR-M4	This study
rkM5	M145:: <i>rk-682</i> in HCR-M5	This study
rkM6	M145:: <i>rk-682</i> in HCR-M6	This study
rkM7	M145:: <i>rk-682</i> in HCR-M7	This study
rkM8	M145:: <i>rk-682</i> in HCR-M8	This study
rkM9	M145:: <i>rk-682</i> in HCR-M9	This study
rkM10	M145:: <i>rk-682</i> in HCR-M10	This study
rkL1	M145:: <i>rk-682</i> in HCR-L1	This study
rkL2	M145:: <i>rk-682</i> in HCR-L2	This study
rkL3	M145:: <i>rk-682</i> in HCR-L3	This study
rkL4	M145:: <i>rk-682</i> in HCR-L4	This study
rkL5	M145:: <i>rk-682</i> in HCR-L5	This study
rkL6	M145:: <i>rk-682</i> in HCR-L6	This study
rkL7	M145:: <i>rk-682</i> in HCR-L7	This study
rkL8	M145:: <i>rk-682</i> in HCR-L8	This study
rkL9	M145:: <i>rk-682</i> in HCR-L9	This study
rkL10	M145:: <i>rk-682</i> in HCR-L10	This study
rkAttB	M145:: <i>rk-682</i>	This study

Table S3. *E. coli* strains used in the study

Strain	Relevant genotype and characteristics	Source
DH10B	<i>LacZ mcrA mcrBC mrr hsd RMS</i>	(15)
ET12567/pUZ8002	<i>rec^F dam dcm Cml^R Str^R Tet^R Km^R</i>	(16)

Table S4. sgRNA information in the study

Name	Sequence (5' to 3')	Target site	Target site location (nt)
sgRNA-M1	ATCTGAACATCGACGCGCCA	M1	4248182
sgRNA-M2	GCGAGGTCACGCCGACGCAC	M2	4368050
sgRNA-M3	CCGGGCGCGCGACCTAACGG	M3	5068174

sgRNA-M4	GGAACCGTGCCAGTGAACAC	M4	3477732
sgRNA-M5	GCACGGGTGGGCGCAGGCGG	M5	2838948
sgRNA-M6	CCACCGCGAGTTCATTCGAA	M6	6083988
sgRNA-M7	TTTACCGTTCCCTTACCCGG	M7	2036981
sgRNA-M8	CTCTTCACAAAATTCTCAGG	M8	883103
sgRNA-M9	TGTTGGGCTCGCCTTCACCG	M9	8158494
sgRNA-M10	CCTTGGCTGCCTGTGTAAA	M10	8621914
sgRNA-L1	AGCACCAGCCCCTGCCCCAG	L1	4203993
sgRNA-L2	GACGACTAGGGCCTGTCCCG	L2	4423541
sgRNA-L3	CATACCCGTCAACGACGAAA	L3	5038644
sgRNA-L4	CACGACTGACGGCGGCCGCC	L4	4903848
sgRNA-L5	GGTTCAGCGGGTTGCGCGG	L5	6058305
sgRNA-L6	TTTACCGTTCCCTTACCCGG	L6	2036981
sgRNA-L7	TCAAGTGGCCTGCATCGAAG	L7	6953647
sgRNA-L8	GCGGCGCGACCTCTGACGAT	L8	8414424
sgRNA-L9	CAACCGATCTTGGAACGTGC	L9	66659
sgRNA-L10	CCTTGGCTGCCTGTGTAAA	L10	8621914

Table S5. Plasmids used in the study

Name	Relevant genotype and characteristics	Source
pWHU2653	<i>ori</i> (ColE1), <i>ori</i> (pIJ101), <i>oriT</i> (RK2), <i>bla</i> , <i>tsr</i> , <i>aac</i> (3)/IV, <i>spcas9</i> , sgRNA cloning cassette, <i>codA</i> (sm)	(17)
pSET152	<i>ori</i> (ColE1), <i>aac</i> (3)/IV, <i>oriT</i> (RK2), <i>ori</i> (ColE1), <i>int</i> , <i>attP</i> (ΦC31), LacZα	(18)
pGAM1	pWHU2653 + sgRNA-M1 + <i>gus</i> (a) + M1-flaning	This study
pGAM2	pWHU2653 + sgRNA-M2 + <i>gus</i> (a) + M2-flaning	This study
pGAM3	pWHU2653 + sgRNA-M3 + <i>gus</i> (a) + M3-flaning	This study
pGAM4	pWHU2653 + sgRNA-M4 + <i>gus</i> (a) + M4-flaning	This study
pGAM5	pWHU2653 + sgRNA-M5 + <i>gus</i> (a) + M5-flaning	This study
pGAM6	pWHU2653 + sgRNA-M6 + <i>gus</i> (a) + M6-flaning	This study
pGAM7	pWHU2653 + sgRNA-M7 + <i>gus</i> (a) + M7-flaning	This study
pGAM8	pWHU2653 + sgRNA-M8 + <i>gus</i> (a) + M8-flaning	This study

pGAM9	pWHU2653 + sgRNA-M9 + <i>gus(a)</i> + M9-flaning	This study
pGAM10	pWHU2653 + sgRNA-M10 + <i>gus(a)</i> + M10-flaning	This study
pGAL1	pWHU2653 + sgRNA-L1 + <i>gus(a)</i> + L1-flaning	This study
pGAL2	pWHU2653 + sgRNA-L2 + <i>gus(a)</i> + L2-flaning	This study
pGAL3	pWHU2653 + sgRNA-L3 + <i>gus(a)</i> + L3-flaning	This study
pGAL4	pWHU2653 + sgRNA-L4 + <i>gus(a)</i> + L4-flaning	This study
pGAL5	pWHU2653 + sgRNA-L5 + <i>gus(a)</i> + L5-flaning	This study
pGAL6	pWHU2653 + sgRNA-L6 + <i>gus(a)</i> + L6-flaning	This study
pGAL7	pWHU2653 + sgRNA-L7 + <i>gus(a)</i> + L7-flaning	This study
pGAL8	pWHU2653 + sgRNA-L8 + <i>gus(a)</i> + L8-flaning	This study
pGAL9	pWHU2653 + sgRNA-L9 + <i>gus(a)</i> + L9-flaning	This study
pGAL10	pWHU2653 + sgRNA-L10 + <i>gus(a)</i> + L10-flaning	This study
pGAattB	pSET152 + <i>gus(a)</i>	This study
pRKM1	pWHU2653 + sgRNA-M1 + <i>rk-682</i> + M1-flaning	This study
pRKM2	pWHU2653 + sgRNA-M2 + <i>rk-682</i> + M2-flaning	This study
pRKM3	pWHU2653 + sgRNA-M3 + <i>rk-682</i> + M3-flaning	This study
pRKM4	pWHU2653 + sgRNA-M4 + <i>rk-682</i> + M4-flaning	This study
pRKM5	pWHU2653 + sgRNA-M5 + <i>rk-682</i> + M5-flaning	This study
pRKM6	pWHU2653 + sgRNA-M6 + <i>rk-682</i> + M6-flaning	This study
pRKM7	pWHU2653 + sgRNA-M7 + <i>rk-682</i> + M7-flaning	This study
pRKM8	pWHU2653 + sgRNA-M8 + <i>rk-682</i> + M8-flaning	This study
pRKM9	pWHU2653 + sgRNA-M9 + <i>rk-682</i> + M9-flaning	This study
pRKM10	pWHU2653 + sgRNA-M10 + <i>rk-682</i> + M10-flaning	This study
pRKL1	pWHU2653 + sgRNA-L1 + <i>rk-682</i> + L1-flaning	This study
pRKL2	pWHU2653 + sgRNA-L2 + <i>rk-682</i> + L2-flaning	This study
pRKL3	pWHU2653 + sgRNA-L3 + <i>rk-682</i> + L3-flaning	This study
pRKL4	pWHU2653 + sgRNA-L4 + <i>rk-682</i> + L4-flaning	This study
pRKL5	pWHU2653 + sgRNA-L5 + <i>rk-682</i> + L5-flaning	This study
pRKL6	pWHU2653 + sgRNA-L6 + <i>rk-682</i> + L6-flaning	This study
pRKL7	pWHU2653 + sgRNA-L7 + <i>rk-682</i> + L7-flaning	This study
pRKL8	pWHU2653 + sgRNA-L8 + <i>rk-682</i> + L8-flaning	This study
pRKL9	pWHU2653 + sgRNA-L9 + <i>rk-682</i> + L9-flaning	This study
pRKL10	pWHU2653 + sgRNA-L10 + <i>rk-682</i> + L10-flaning	This study
pRKattB	pSET152 + <i>rk-682</i>	This study

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